Centralized IT Infrastructure Setup for a Growing SME

727823TUIT223 – SIVAKUMAR R

Problem Statement:

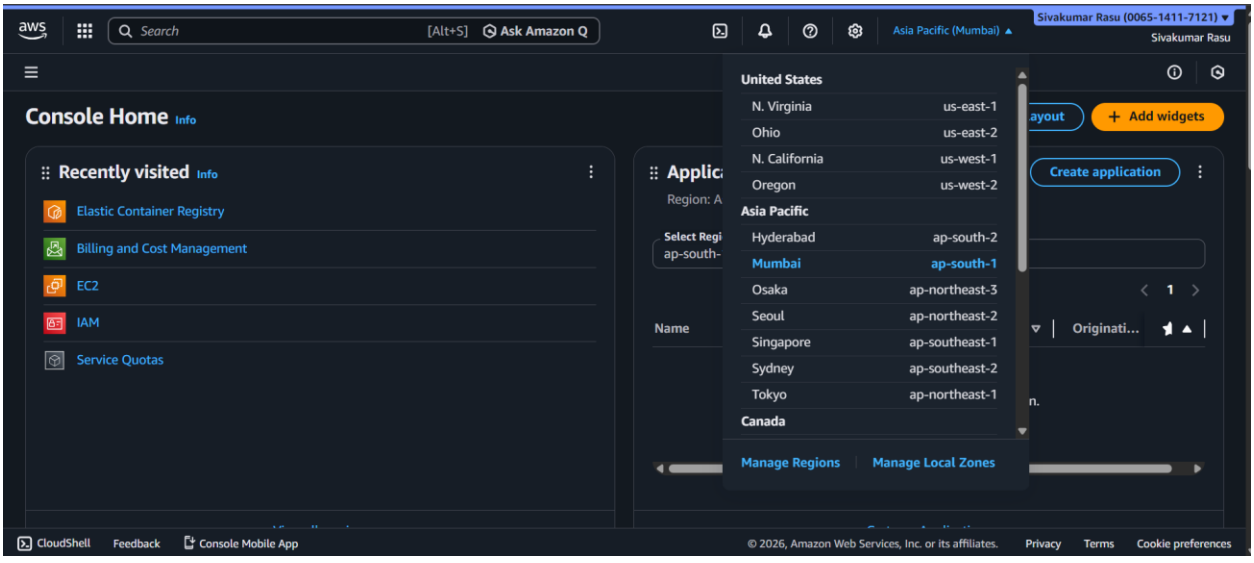
A small enterprise with 25 employees is facing frequent system downtime, unstructured data storage, and lack of remote access. All applications and files currently run on individual employee systems with no centralized control. The IT head wants to move to a cloud-based infrastructure that is stable, manageable, and scalable without increasing operational complexity. The challenge is that the company has no prior cloud experience, and the new intern must design and implement a basic but production-ready cloud infrastructure.

Project Description:

- A virtualized environment using a hypervisor
- A cloud-hosted virtual machine acting as a central server
- Proper CPU, memory, and storage allocation
- Secure access for administrators
- Backup and recovery capability

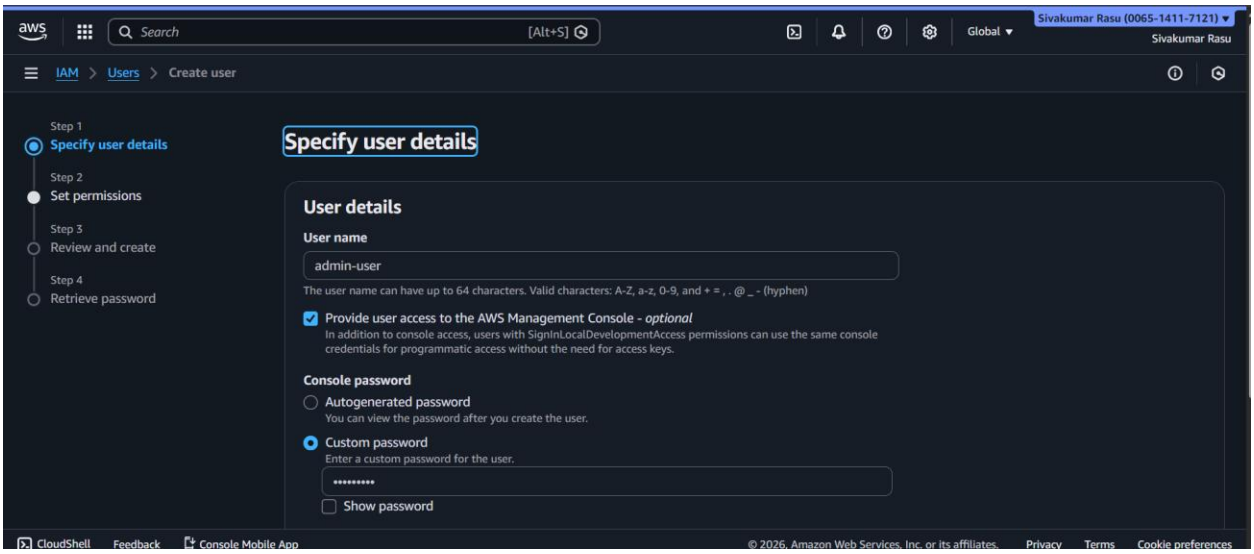
1.Account Creation & Region Selection

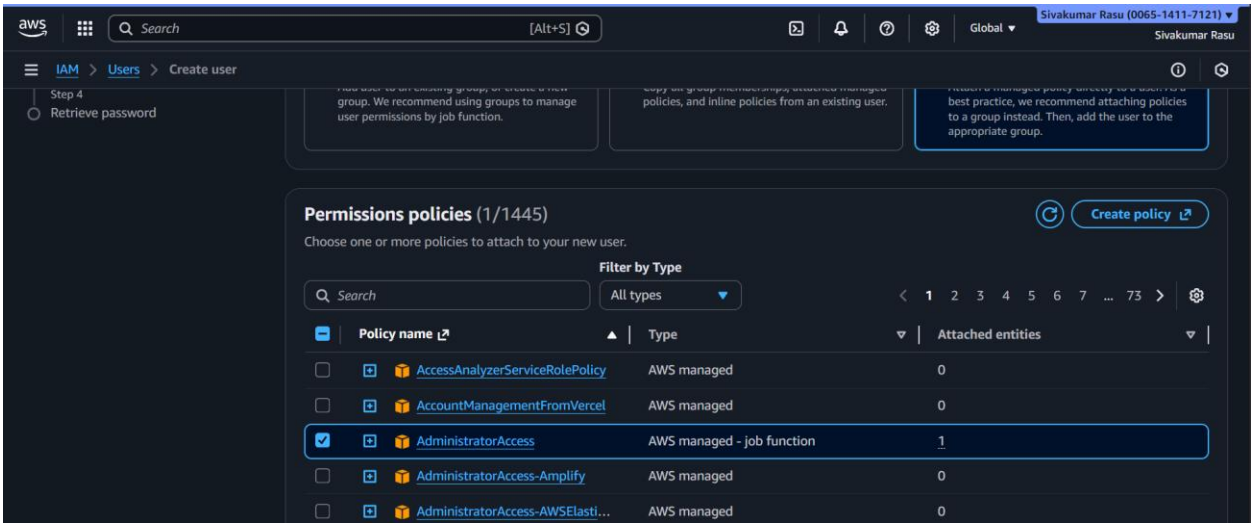
Created an AWS account and selected region for implementing Centralized cloud IT infrastructure for Small Enterprises.

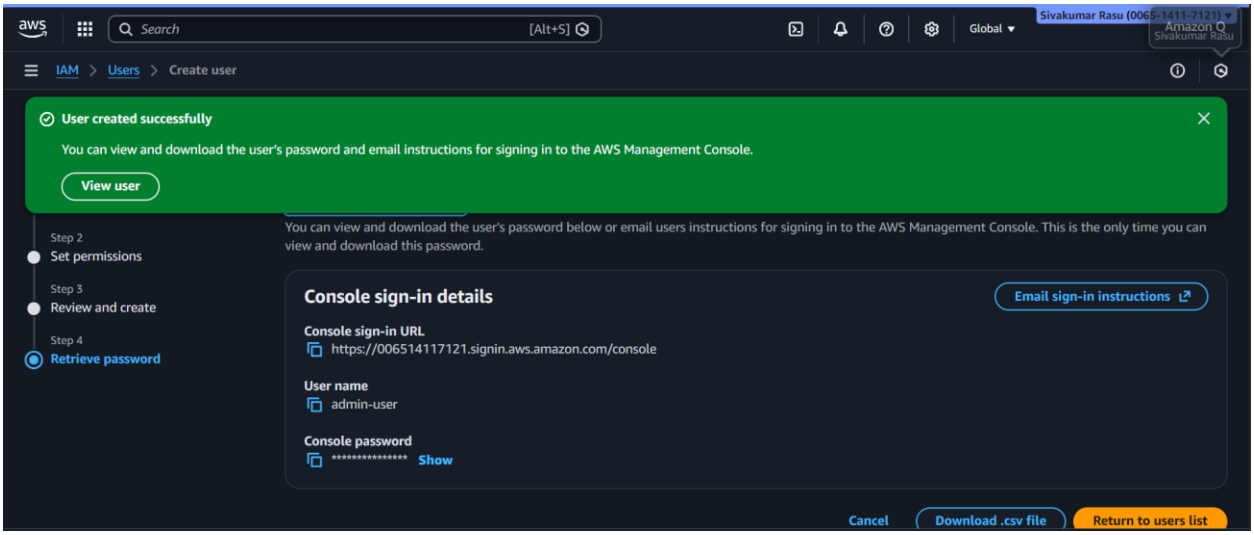


2.Secure admin access setup

Created IAM role for admin user who handles all the tasks, employees and other admin level operations.

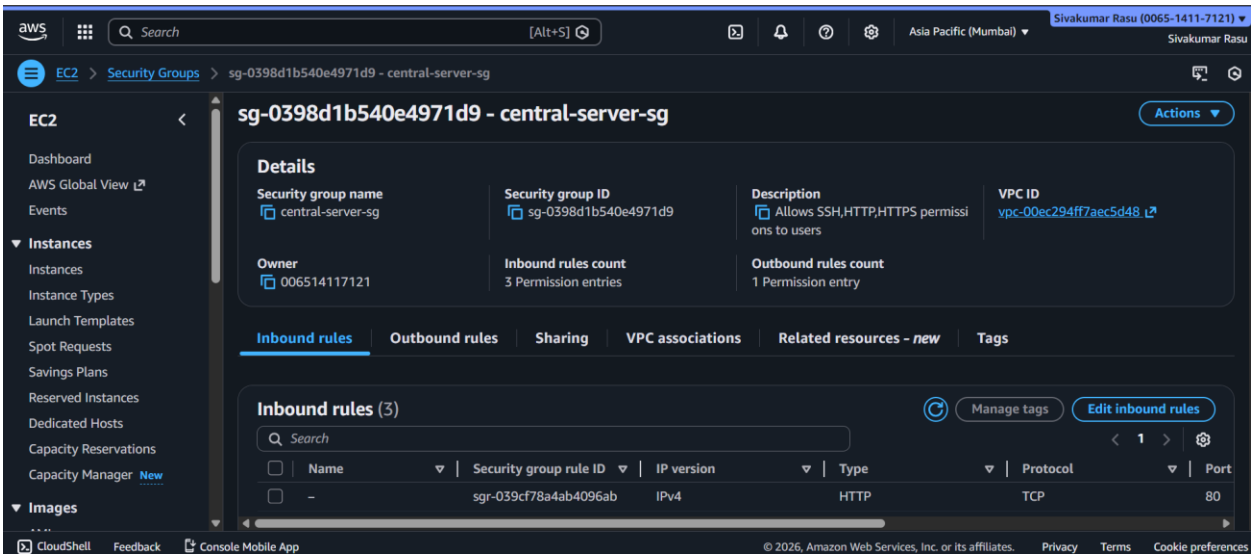






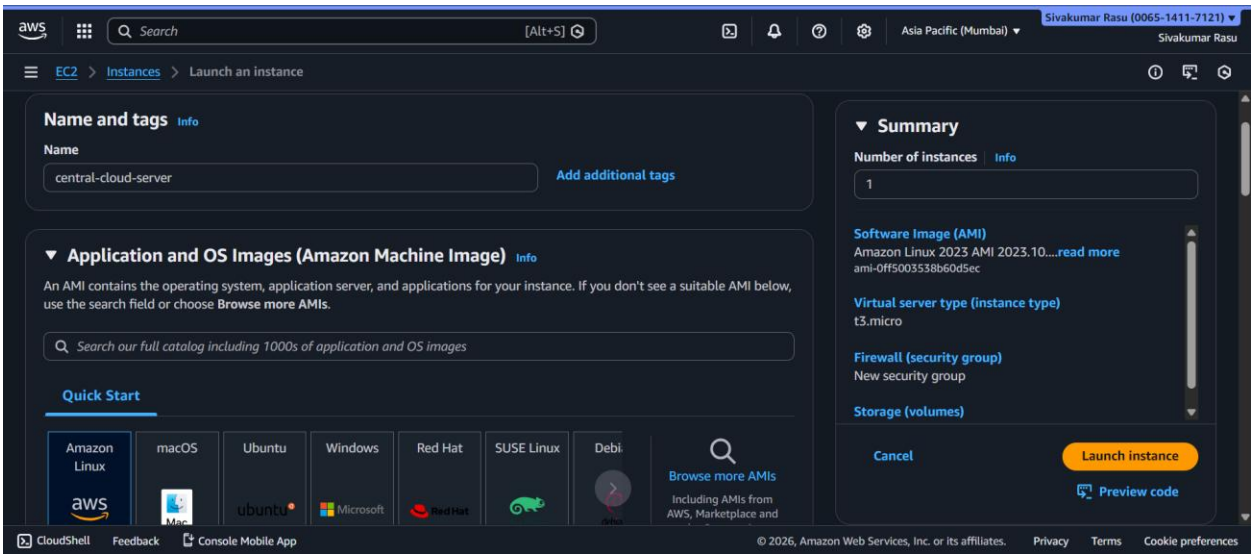
3.Creating Security Group

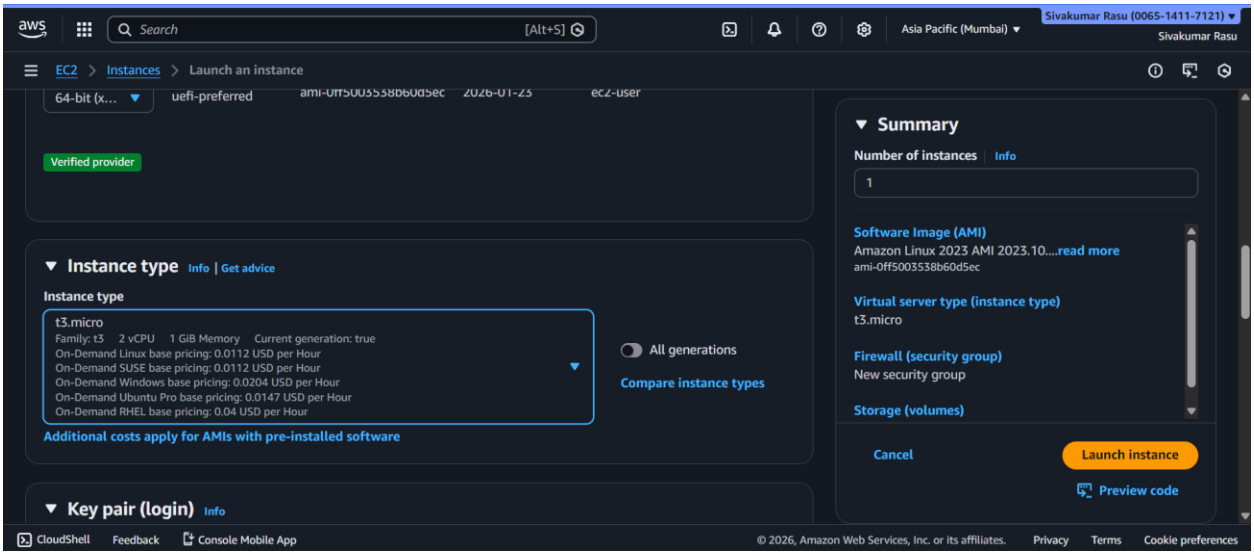
Created security group for the servers that allowing ports like SSH, HTTP, HTTPS to the users who all are accessing the resources via the central server.

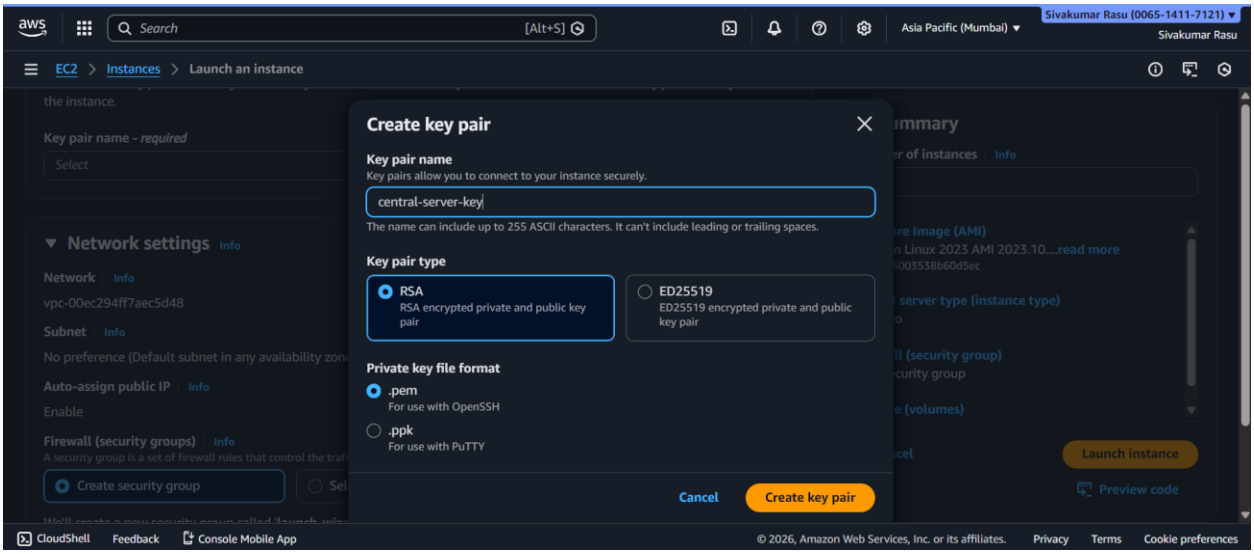


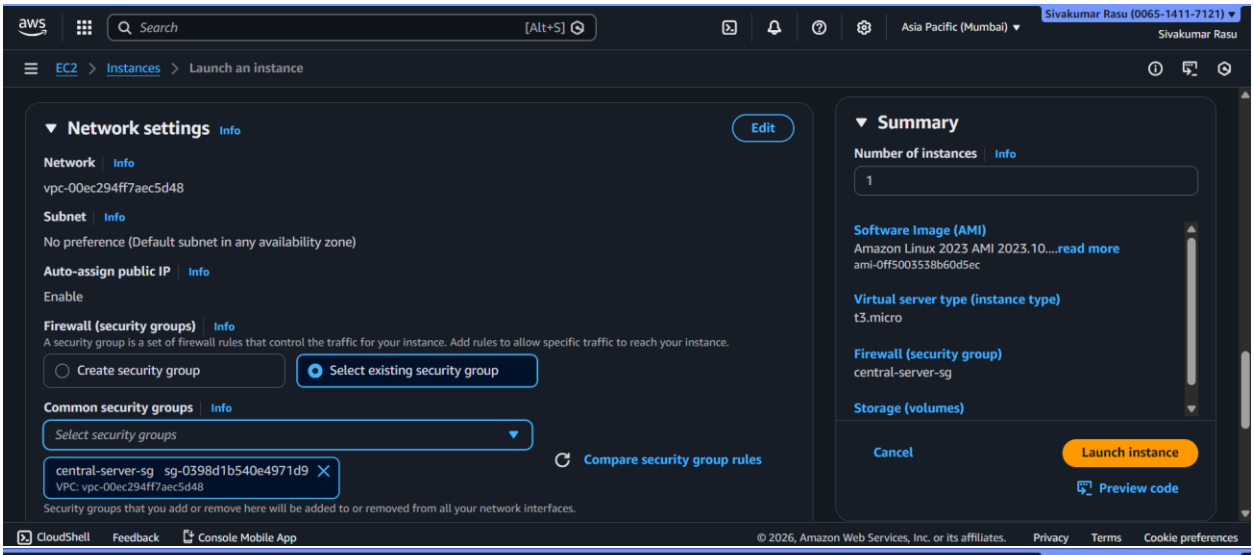
4.Virtual Server Creation

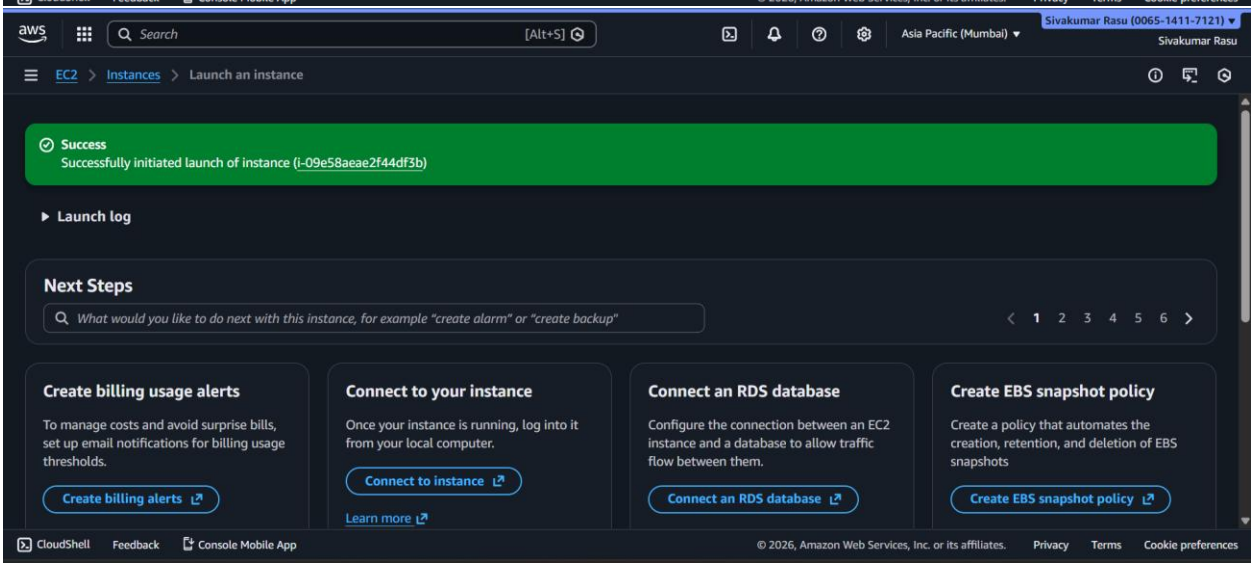
Created a centralized cloud server using AWS EC2 for the company to provide resources across the network.





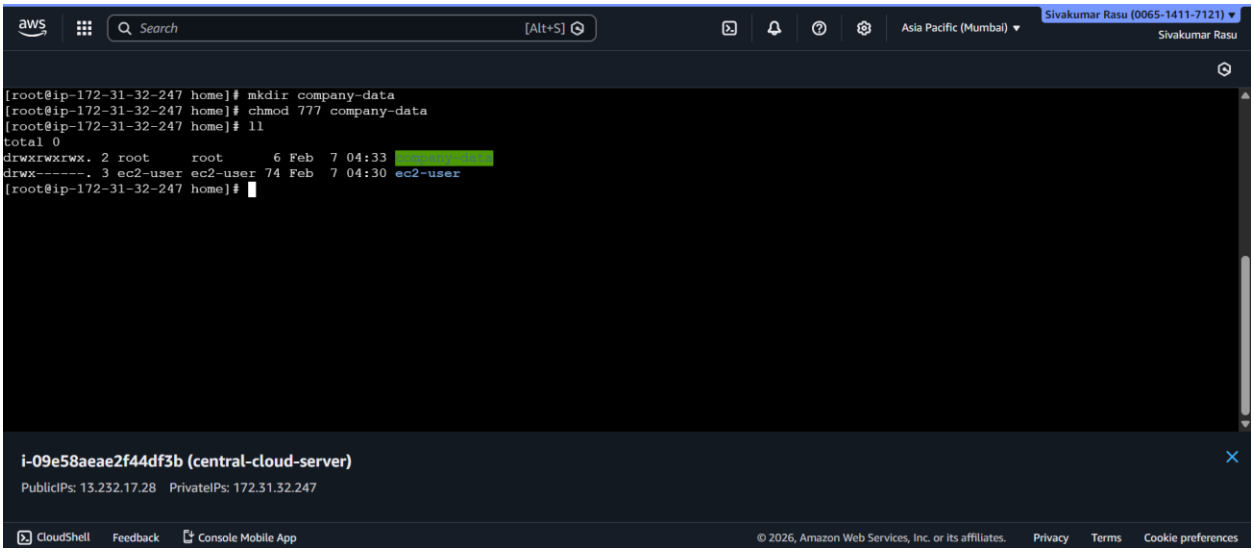

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EC2 > Instances > Launch an instance
Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
central-server-key
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
Cancel
Create key pair
Network settings Info
Network Info
vpc-00ec294ff7aec5d48
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic
Create security group
Launch instance
Preview code
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EC2 > Instances > Launch an instance
Network settings Info
Edit
Network Info
vpc-00ec294ff7aec5d48
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
central-server-sg sg-0398d1b540e4971d9
VPC: vpc-00ec294ff7aec5d48
Compare security group rules
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-0ff5003538b60d5ec
Virtual server type (instance type)
t3.micro
Firewall (security group)
central-server-sg
Storage (volumes)
Cancel
Launch instance
Preview code
CloudShell
Feedback
Console Mobile App
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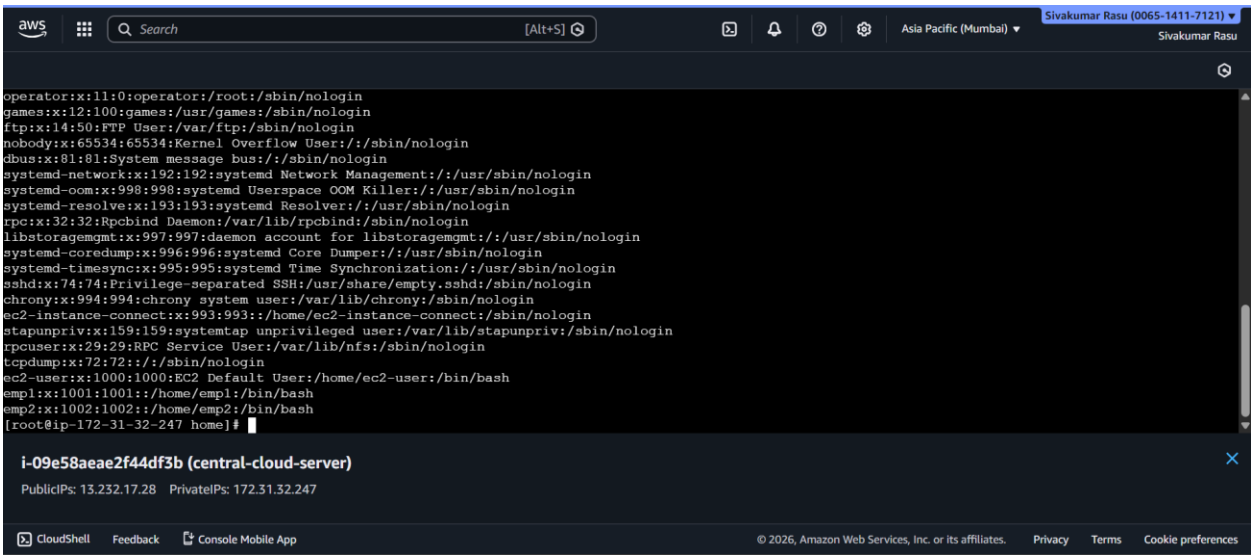

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EC2 > Instances > Launch an instance
Success
Successfully initiated launch of instance (i-09e58aeae2f44df3b)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
1 2 3 4 5 6
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
CloudShell
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5.Created Central File Storage

Created a centralized file storage with public access. Every employee from the company can read and write data to the file storage at the path /home/company-data

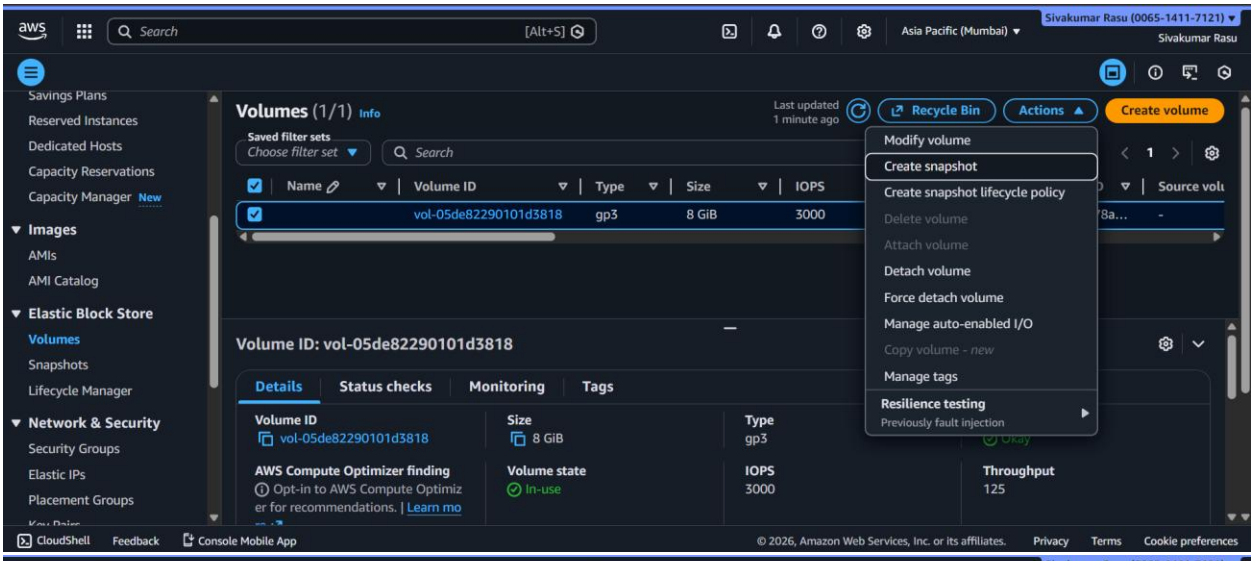


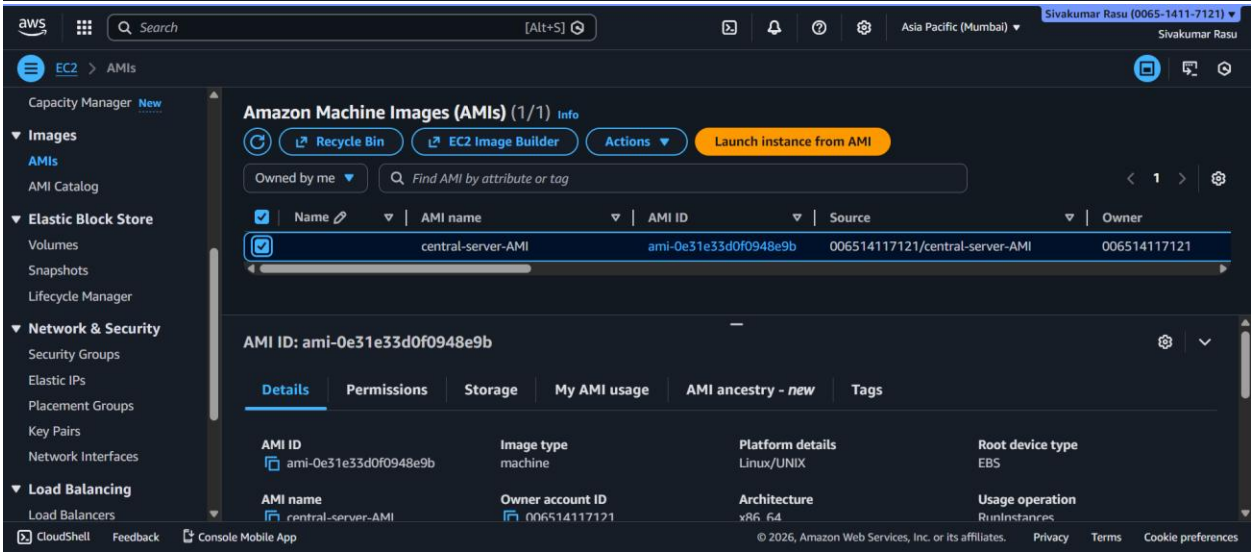
Created employee accounts that who all are working at the company

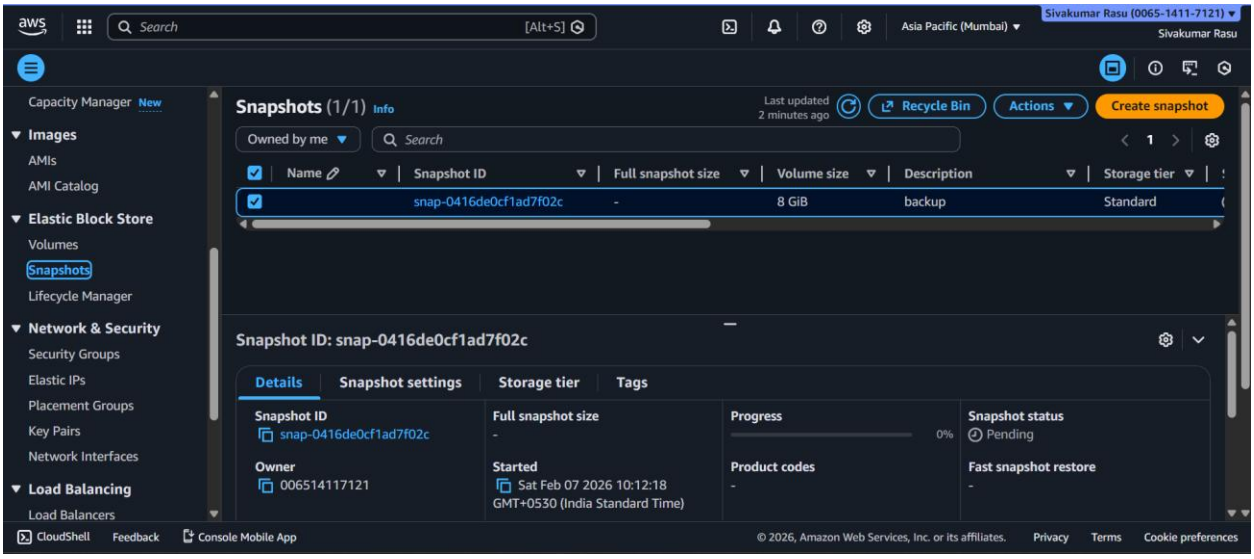


6.Backup & Recovery

Created AMI snapshots and images for maintaining the backup of the central-server for disaster recovery







7.Archietecture Diagram

